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Gainesville, FL 32611
April 5, 2026

Dr. Kurt Schulze
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Dear Dr. Schulze:

The objective of this experiment is to characterize the thermal performance of a packed column cooling tower through analysis of heat and mass transfer between air and water streams. This experiment evaluates the variation in air humidity ratio and relative humidity between the inlet and exit conditions, predicts the exit water temperature using a first law energy balance, and determines the rate of heat dissipation within the system. The measured results are compared to analytical predictions to assess the accuracy of the governing thermodynamic relationships and the effectiveness of the cooling tower under varying water flow rates and heating conditions.

The analysis was based on the principles of heat and mass transfer in an air-water vapor system undergoing evaporative cooling within a packed column. As unsaturated air flowed counter to the falling water, evaporation occurred, increasing the moisture content of the air while reducing the water temperature, consistent with conservation of mass and energy. The system was assumed to operate at steady state and constant pressure with negligible heat transfer to surroundings or changes in kinetic and potential energy. The thermodynamic state of the air at the inlet and outlet was determined using measured dry and wet bulb temperatures, from which humidity ratio and relative humidity were obtained with a series of interpolations from the psychometric chart. Air and water vapor were treated as ideal gases and thus the air density was solved using the ideal gas law and was then equated with the outlet humidity ratio to find the vapor density. The mass flow rate of air was obtained by equating air density with the pressure differential measurement, and then the outlet water mass flow rate was found by equating the known inlet water mass flow rate, the mass flow rate of air, and the humidity ratio across the inlet and outlet. The steady-flow energy equation, expressed in terms of enthalpy per unit mass of dry air, related inlet and outlet conditions to predict the exit water temperature. Similarly, heat dissipation was determined by applying the steady-flow energy balance to both the air and water streams, where the rate of heat transfer was calculated from the change in enthalpy between inlet and outlet conditions.

The experiment was conducted using a bench-scale packed column cooling tower with multiple temperature, pressure and flow measurement devices to characterize both air and water streams. The apparatus included six thermocouples positioned to measure inlet and outlet conditions of both phases, with the paired wet and dry bulb sensors located before and after the column to capture the air properties. Water flow was supplied from a heated reservoir and distributed at the top of the column, while air was driven upward through the packing by a blower with airflow rate determined from a differential pressure measurement across an orifice. Additional instrumentation included a rotameter to measure the water flow rate at three distinct settings, along with a monitored heat power input that was set to both a high and low level for each flow condition, resulting in six test cases. Data was collected after the system had reached steady operating conditions at each combination of water flow rate and heater setting, with temperature, pressure and flow measurements recorded for subsequent analysis. Precautions had been taken to ensure measurement accuracy, including allowing sufficient time to achieve steady state, maintaining saturation of the wet bulb wicks, and using distilled water to minimize scaling effect. Safety practices such as wearing eye protection were followed by the team.

The objectives of characterizing the air-water vapor properties and evaluating the cooling tower performance were successfully addressed through comparison of inlet and outlet conditions and predicted versus measured values. The humidity ratio had increased consistently across all test cases with inlet values ranging from 0.00956 to 0.01004 kg/kg dry air and outlet values ranging from 0.01298 to 0.01773 kg/kg dry air, indicating effective moisture transfer to the air stream. Similarly, relative humidity increased from about 64-66% at the inlet to 96-100% at the outlet, confirming that the air approached saturation as expected. The predicted exit water temperatures had shown reasonable agreement with measured values, with differences ranging from 0.44 °C to 2.49 °C, although larger deviations were observed at higher heating inputs. Heat dissipation predictions consistently exceeded measured values, with differences ranging from 0.07 kW to 0.66 kW, suggesting systematic overprediction by the model. These discrepancies were likely attributable to heat losses to the surroundings, measurement uncertainty, and ideal behavior assumptions. Despite these deviations, the results demonstrated consistent trends and agreement with theoretical expectations, indicating that the thermodynamic analysis provided an effective representation of cooling tower performance.

Sincerely,

Bryce Addeo | Sean Muller | Matthew Strogach

"On my honor, I have neither given nor received unauthorized aid in doing this assignment.

Table 1: The table below shows results for the inlet and outlet humidity ratio, ω_{in} and ω_{out}, and the inlet and outlet relative humidity, φ_{in} and φ_{out}, for all six test cases. The first three test cases involved low heater input (0.37 kW) with flow rates of 20 g/s, 30 g/s, and 40 g/s respectively, while the latter three test cases involved higher heater input (1.228 kW) with the same three flow rate settings. The presented results for both humidity ratio and relative humidity were found using the same method that involved plotting on the psychometric chart and interpolation. The bounds of wet and dry bulb inlet and outlet temperatures were plotted, and the corresponding humidity ratio and relative humidity values were found at their intersection. In all, eight values of inlet humidity ratios and relative humidity were found at different combinations of low and high wet and dry bulb bounds, and then a series of three interpolations were conducted to determine the final inlet humidity ratio and relative humidity for each test case. The same exact process was then repeated using outlet wet and dry bulb temperatures to solve the outlet humidity ratio and relative humidity. The reason why three interpolations were required was because it was not possible to plot decimal place temperature values with perfect precision, nor was it possible to perfectly measure in between the bounds of the humidity ratio and relative humidity lines and curves. Therefore, multiple interpolations were used to ensure results were as accurate as possible. The values shown below demonstrate a consistent increase in both humidity ratio and relative humidity between the inlet and outlet air streams across all test cases, confirming an evaporative cooling within the tower. The inlet humidity ratio remained relatively constant, ranging narrowly from 0.00956 to 0.01004 kg/kg dry air, with higher values corresponding to increased water flow rates and heater inputs. An outlier to this trend was test case 1 at the inlet because it had the third highest humidity ratio value despite corresponding to the lowest heater power input and lowest flow rate setting. In contrast, the outlet humidity ratio increased significantly with operating conditions, ranging from 0.01298 to 0.01773 kg/kg dry air, with higher values corresponding to increased water flow rates and heating inputs as theoretically expected. This trend reflected enhanced liquid-air contact, allowing more moisture to be transferred into the air stream. A similar pattern was observed for relative humidity, where inlet values remained in between approximately 63.90% and 66.4% while outlet values approached saturation, ranging from 96.6% to 100%. The outlet air reaching near-saturation indicated that the cooling tower operated close to its thermodynamic limit, as the driving force for evaporation diminished when the air became saturated. Additionally, the increase in outlet humidity with higher heating power suggested that elevated water temperatures promoted greater evaporation rates, while increased water flow provided a larger surface area for mass transfer.

Test	ω_{in} **(kg/kg dry air)**	ω_{out} **(kg/kg dry air)**	φ_{in} **(%)**	φ_{out} **(%)**
1	0.00988	0.01298	65.60	96.64
2	0.00956	0.01364	63.90	98.27
3	0.00968	0.01372	64.69	97.71
4	0.00976	0.01602	64.86	98.60
5	0.01000	0.01770	66.42	100.0
6	0.01004	0.01773	65.90	98.56

Table 2: The table below showcases results for the predicted exit water temperature, $T_{exit,predicted}$, the actual measured exit water temperature, $T_{exit,actual}$, the difference between those temperature values, ΔT, the predicted heat dissipation, $Q_{predicted}$, the actual heat dissipation, Q_{actual}, and the difference between those heat dissipation values, ΔQ. The density and mass flow rate of both air and water streams were used to determine outlet enthalpies that were then interpolated with corresponding temperatures to solve the predicted exit water temperatures. The actual exit water temperatures were measured by the experimental apparatus, and the difference was found by subtracting the predicted values from the measured value for each test case. Similarly, heat dissipation was predicted theoretically by equating mass flow rate and enthalpies, the actual heat dissipation was the measured heating power input during the experiment, and the difference was the predicted value subtracted from the actual value. The predicted exit water temperatures had consistently underestimated the measured values with differences ranging from 0.44 °C to 2.49 °C. The smallest deviation occurred at the lower heating power input, while significantly larger deviations were observed at higher heating conditions, indicating the model accuracy decreased as thermal loading increased. This trend suggested that assumptions such as negligible heat loss to surroundings and idealized energy transfer became less valid at elevated temperatures, where convective losses were more pronounced. On the other hand, a similar but opposite pattern was observed in the heat dissipation results, where the predicted values consistently exceeded the measured heater input, with the difference ranging from -0.07 kW to -0.66 kW. The increasing magnitude of this discrepancy at higher heater input indicated a systematic overprediction of heat transfer by the model. This behavior was likely due to cumulative uncertainty in air mass flow rate determination, enthalpy values obtained from the psychometric chart, and the assumption that all energy exchange occurred strictly between the air and water streams. However, the fourth test case presented an outlier in the results where the predicted heat dissipation value was nearly equal to the measured value even with the higher heating power input. Despite these discrepancies, the predicted and measured results followed consistent trends, with both exit water temperature and heat dissipation increasing with higher heating power and flow rate.

Test	$T_{exit,predicted}$ **(°C)**	$T_{exit,actual}$ **(°C)**	ΔT **(°C)**	$Q_{predicted}$ **(kW)**	Q_{actual} **(kW)**	ΔQ **(kW)**
1	17.59	18.50	0.91	0.50	0.37	-0.13
2	18.66	19.10	0.44	0.57	0.37	-0.20
3	18.45	19.30	0.85	0.74	0.37	-0.37
4	18.11	20.60	2.49	1.30	1.228	-0.07
5	18.72	21.20	2.48	1.82	1.228	-0.59
6	20.86	23.00	2.14	1.89	1.228	-0.66

The sample calculations for humidity ratio and relative humidity both use linear interpolation to achieve precise results. This was necessary because the measured temperatures didn't often line up perfectly with the whole number temperatures denoted on the chart. To calculate properties more closely associated with our temperature measurements, linear interpolation was used. The general linear interpolation formula is shown below, where the measured value x falls between upper bound, x_2, and lower bound, x_1. The values y_2 and y_1 are the output associated with each input boundary. The final output, y, represents the linearly interpolated output value associated with the input, x.

$$y = y_1 + \frac{x - x_1}{x_2 - x_1}(y_2 - y_1)\ No.3, p.72\ Bergman\ et\ al.\ [1] \tag{1}$$

This sample calculation solves for the humidity ratio at the inlet, ω_{in}, with a psychometric chart and linear interpolation, using the inlet data from test 2, where the heater was set to 0.5 kW and the water flow rate was set to 30 g/s. The humidity ratio boundaries, ω_{low} and ω_{high}, were first interpolated linearly with respect to T_1, where ω_{low} had a lower boundary of $\omega_{low,low}$ (humidity ratio at the intersection of low dry bulb bound and low wet bulb bound) and an upper boundary of $\omega_{high,low}$ (humidity ratio at the intersection of high dry bulb bound and low wet bulb bound). Similarly, ω_{high} had a lower boundary of $\omega_{low,high}$ (humidity ratio at the intersection of low dry bulb bound and high wet bulb bound) and an upper boundary of $\omega_{high,high}$ (humidity ratio at the intersection of high dry bulb bound and high wet bulb bound). By interpolating with respect to T_1, the offset of the dry bulb temperature from the whole number boundaries on the psychometric chart could be accounted for in a linear manner, resulting in humidity ratio boundaries that better represent the properties at the measured dry bulb temperature.

$$\omega_{low} = \omega_{low,low} + \frac{T_1 - T_{1,low}}{T_{1,high} - T_{1,low}}\left(\omega_{high,low} - \omega_{low,low}\right)$$

$$\omega_{low} = 0.0098 + \frac{20.6\ ℃ - 20\ ℃}{21\ ℃ - 20\ ℃}(0.0094 - 0.0098) = 0.00956$$

$$\omega_{high} = \omega_{low,high} + \frac{T_1 - T_{1,low}}{T_{1,high} - T_{1,low}}\left(\omega_{high,high} - \omega_{low,high}\right)$$

$$\omega_{high} = 0.011 + \frac{20.6\ ℃ - 20\ ℃}{21\ ℃ - 20\ ℃}(0.0106 - 0.011) = 0.01076$$

The humidity ratio at the inlet, ω_{in}, was then calculated through linear interpolation with respect to the inlet wet bulb temperature, T_2. The humidity ratio boundaries, ω_{low} and ω_{high}, already linearly interpolated with respect to the dry bulb temperature, T_1, were then used as the lower and upper boundaries for the linear interpolation with respect to the wet bulb temperature. By interpolating with respect to T_2, the offset of the wet bulb temperature from the whole number boundaries on the psychometric chart could be accounted for in a linear manner, resulting in an inlet humidity ratio that better represents the properties at the measured wet bulb temperature.

$$\omega_{in} = \omega_{low} + \frac{T_2 - T_{2,low}}{T_{2,high} - T_{2,low}}\left(\omega_{high} - \omega_{low}\right)$$

$$\omega_{in} = 0.00956 + \frac{16\ ℃ - 16\ ℃}{17\ ℃ - 16\ ℃}(0.01076 - 0.00956)$$

$$\therefore \omega_{in} = 0.00956$$

The process above was then repeated with the outlet dry bulb temperature, T_3, and the outlet wet bulb temperature, T_4, used in place of the inlet dry bulb temperature, T_1, and inlet wet bulb temperature, T_2, respectively. The resulting humidity ratio, ω_{out}, reflects the conditions at the outlet.

This sample calculation solves for the relative humidity at the inlet, φ_{in}, with a psychometric chart and linear interpolation, using the inlet data from test 2, where the heater was set to 0.5 kW and the water flow rate was set to 30 g/s. The relative humidity boundaries, φ_{low} and φ_{hig} , were first interpolated linearly with respect to T_1, where φ_{low} had a lower boundary of $\varphi_{low,low}$ and an upper boundary of $\varphi_{high,low}$. Similarly, φ_{high} had a lower boundary of $\varphi_{low,high}$ and an upper boundary of $\varphi_{high,hig}$. By interpolating with respect to T_1, the offset of the dry bulb temperature from the whole number boundaries on the psychometric chart could be accounted for in a linear manner, resulting in relative humidity boundaries that better represent the properties at the measured dry bulb temperature.

$$\varphi_{low} = \varphi_{low,low} + \frac{T_1 - T_{1,low}}{T_{1,high} - T_{1,low}} * \left(\varphi_{high,low} - \varphi_{low,low}\right)$$

$$\varphi_{low} = 67.5\ \% + \frac{20.6\ ^{\circ}\mathrm{C} - 20\ ^{\circ}\mathrm{C}}{21\ ^{\circ}\mathrm{C} - 20\ ^{\circ}\mathrm{C}} * (61.5\ \% - 67.5\ \%) = 63.9\ \%$$

$$\varphi_{high} = \varphi_{low,high} + \frac{T_1 - T_{1,low}}{T_{1,high} - T_{1,low}} * \left(\varphi_{high,high} - \varphi_{low,high}\right)$$

$$\varphi_{high} = 76\ \% + \frac{20.6\ ^{\circ}\mathrm{C} - 20\ ^{\circ}\mathrm{C}}{21\ ^{\circ}\mathrm{C} - 20\ ^{\circ}\mathrm{C}} * (69\ \% - 76\ \%) = 71.8\ \%$$

The relative humidity at the inlet, φ_{in}, was then calculated through linear interpolation with respect to the inlet wet bulb temperature, T_2. The relative humidity boundaries, φ_{low} and φ_{high}, already linearly interpolated with respect to the dry bulb temperature, T_1, were then used as the lower and upper boundaries for the linear interpolation with respect to the wet bulb temperature. By interpolating with respect to T_2, the offset of the wet bulb temperature from the whole number boundaries on the psychometric chart could be accounted for in a linear manner, resulting in an inlet relative humidity that better represents the properties at the measured wet bulb temperature.

$$\varphi_{in} = \varphi_{low} + \frac{T_2 - T_{2,low}}{T_{2,high} - T_{2,low}} * \left(\varphi_{high} - \varphi_{low}\right)$$

$$\varphi_{in} = 63.9\ \% + \frac{16\ ^{\circ}\mathrm{C} - 16\ ^{\circ}\mathrm{C}}{17\ ^{\circ}\mathrm{C} - 16\ ^{\circ}\mathrm{C}} * (71.8\ \% - 63.9\ \%)$$

$$\therefore \varphi_{in} = 63.9\ \%$$

The process above was then repeated with the outlet dry bulb temperature, T_3, and the outlet wet bulb temperature, T_4, used in place of the inlet dry bulb temperature, T_1, and inlet wet bulb temperature, T_2, respectively. The resulting relative humidity, φ_{out}, reflects the conditions at the outlet.

The following sample calculation solves for the density of the dry air, ρ_{air}, which does not include any water vapor. This variable explains how much dry air mass is packed into a given volume. It is dependent on the atmospheric pressure, P_{atm}, the gas constant for air, R_{air}, and the inlet dry bulb temperature, T_1. The inlet dry bulb temperature is the actual temperature of the air entering the cooling tower which is measured without considering moisture. Then, the mixture density, ρ_{mix}, can be found which is a mix of air and water vapor. It is used in calculating total air vapor flow and is dependent on the density of the dry air, ρ_{air}, and the humidity ratio of the air leaving the cooling tower, ω_{out}. Both sample calculations were solved for test case 2.

$$\rho_{air} = \frac{P_{atm}}{R_{air} * T_1} \; p.18 \; Schulze \; [2] \tag{2}$$

$$\rho_{mix} = \rho_{air} * (1 + \omega_{out}) \; pp.7 - 12 \; Schulze \; [3] \tag{3}$$

Furthermore, it is then possible to calculate the mass flowrate of the dry air, $\dot{m}_{air}$, which describes how much dry air is flowing through the cooling tower per second. It is dependent on the mixture density, ρ_{mix}, as well as the pressure drop of the cooling tower, ΔP, which was a given. The mass flowrate of the water leaving the cooling tower, $\dot{m}_{w,out}$, could also be found, which represents how much water is leaving the cooling tower per second. It is based on the mass flowrate of the dry air, $\dot{m}_{air}$, the mass flowrate of the water entering the tower, $\dot{m}_{w,in}$, and the humidity ratio of the air entering the cooling tower, ω_{in}, and the humidity ratio of the air leaving the cooling tower, ω_{out}. The mass flowrate of the water entering the tower was a set constant depending on the test case. This was also done for test case 2.

$$\dot{m}_{air} = 0.0137 * \sqrt{\rho_{mix} * \Delta P} \; Eqn. 8.57, p.282 \; Fox \; and \; McDonald \; [4] \tag{4}$$

$$\dot{m}_{w,out} = \dot{m}_{w,in} + \dot{m}_{air} * (\omega_{in} - \omega_{out}) \; p.21 \; Schulze \; [5] \tag{5}$$

Using the derived mass flowrates, the specific enthalpy of the water leaving the cooling tower, $h_{w,out}$, could be found which indicates the amount of energy stored in each kilogram of water based on its outlet temperature. It provides a much better understanding of how cooling happens. This is dependent on the mass flowrate of the water entering the tower, $\dot{m}_{w,in}$, the mass flowrate of the dry air, $\dot{m}_{air}$, the mass flowrate of the water leaving the cooling tower, $\dot{m}_{w,out}$, the enthalpy of the air entering the tower, h_{in}, and the enthalpy of the air leaving the tower, h_{out}. Both of those enthalpy values were found from a Psychometric Chart using the wet and dry bulb temperature values [6]. The inlet enthalpy was found on the chart by using the inlet dry bulb temperature, T_1, and the inlet wet bulb temperature, T_2, and finding their intersection point. The outlet enthalpy was found on the chart by using the outlet dry bulb temperature, T_3, and the outlet wet bulb temperature, T_4, and finding their intersection point. The specific enthalpy of the water leaving the cooling tower is also dependent on the enthalpy of the incoming water, $h_{w,in}$. This value was found through an interpolation of the enthalpy value of the temperature of the hot water entering the cooling tower, T_5 [1].

$$h_{w,out} = \frac{\dot{m}_{air} * (h_{in} - h_{out}) + (\dot{m}_{w,in} * h_{w,in})}{\dot{m}_{w,out}} \; p.10 \; Schulze \; [7] \tag{6}$$

Finally, the outlet water temperature, $T_{predicted,out}$, and the difference between the measured temperature, $T_{measured}$, and predicted temperature, $T_{predicted,out}$. This temperature difference. ΔT, how far the calculated temperature is from the measured temperature. Essentially, the predicted temperature is the expected temperature of the water at the exit based on how much heat should be removed by the air. It is dependent on the water leaving the cooling tower, $h_{w,out}$, the low bound temperature of the water entering the cooling tower, $T_{5,low}$, the high bound temperature of the water entering the cooling tower, $T_{5,high}$, the enthalpy of the outlet air, h_{high}, and the enthalpy of the inlet air, h_{low}. The outlet enthalpy comes from the corresponding value to the higher bound temperature of the cooling tower, and the inlet enthalpy comes from the corresponding value to the lower bound temperature of the cooling tower. The measured temperature in calculating the temperature difference comes from the recorded temperature of the water leaving the cooling tower, T_6.

$$T_{predicted,out} = T_{5,low} + \frac{(h_{w,out} - h_{low})}{(h_{high} - h_{low})} * \left(T_{5,high} - T_{5,low}\right) p.10 \; Schulze \; [7] \tag{7}$$

$$\Delta T = T_{measured} - T_{predicted,out} \; p.10 \; Schulze \; [7] \tag{8}$$

The sample calculation for all these values was completed for test case 2 with a heating power of 0.37 kW, a water flow rate of 30 g/s, and an air pressure differential of 30 mm H2O.

$$P_{atm} = 101325 \; Pa$$

$$R_{air} = 287 \; \frac{J}{kg * K}$$

$$T_1 = 20.6 \; °C + 273.15 = 293.75 \; K$$

$$\rho_{air} = \frac{101325 \; Pa}{287 \; \frac{J}{kg * K} * 293.75 \; K} = \frac{101325 \; \frac{kg}{m * \cancel{s^2}}}{287 \; \frac{m^2}{\cancel{s^2 * K}} * 293.75 \; \cancel{K}}$$

$$\therefore \; \rho_{air} = 1.2019 \; \frac{kg}{m^3}$$

$$\omega_{out} = 0.01364$$

$$\rho_{mix} = 1.2019 \; \frac{kg}{m^3} * (1 + 0.01364)$$

$$\therefore \; \rho_{mix} = 1.2183 \; \frac{kg}{m^3}$$

$$\Delta P = 30 \; mmH2O$$

$$\dot{m}_{air} = 0.0137 * \sqrt{1.2183\ \frac{kg}{m^3} *\ 30\ mmH2O} =\ 0.0137 \frac{\sqrt{kg} * \sqrt{\cancel{m^3}}}{s * \sqrt{\cancel{mmH_2O}}} * \sqrt{1.2183\ \frac{kg}{\cancel{m^3}} *\ 30\ \cancel{mmH_2O}}$$

$$\therefore\ \dot{m}_{air} = 0.08282\ \frac{kg}{s}$$

$$\dot{m}_{w,in} = 0.03\ \frac{kg}{s}$$

$$\omega_{in} = 0.00956$$

$$\dot{m}_{w,out} =\ 0.03\ \frac{kg}{s} +\ 0.08282\ \frac{kg}{s} * (0.00956 - 0.01364)$$

$$\therefore\ \dot{m}_{w,out} = 0.02966\ \frac{kg}{s}$$

$$h_{in} = 65\ \frac{kJ}{kg}$$

$$h_{out} = 71.9\ \frac{kJ}{kg}$$

$$h_{w,in} = 96.39\ \frac{kJ}{kg}$$

$$h_{w,out} = \frac{0.08282\ \frac{\cancel{kg}}{\cancel{s}} * \left(65\ \frac{kJ}{\cancel{kg}} - 71.9\ \frac{kJ}{\cancel{kg}}\right) + \left(0.03\ \frac{\cancel{kg}}{\cancel{s}} * 96.39\ \frac{kJ}{\cancel{kg}}\right)}{0.02966\ \frac{kg}{\cancel{s}}}$$

$$\therefore\ h_{w,out} = 78.22\ \frac{kJ}{kg}$$

$$T_{5,low} = 18\ °C$$

$$h_{high} = 79.65\ \frac{kJ}{kg}$$

$$h_{low} = 75.47\ \frac{kJ}{kg}$$

$$T_{5,high} =\ 19\ °C$$

$$T_{predicted,out} =\ 18\ °C + \frac{\left(78.22\ \frac{\cancel{kJ}}{\cancel{kg}} - 75.47\ \frac{\cancel{kJ}}{\cancel{kg}}\right)}{\left(79.65\ \frac{\cancel{kJ}}{\cancel{kg}} - 75.47\ \frac{\cancel{kJ}}{\cancel{kg}}\right)} * (19\ °C -\ 18\ °C)$$

$$\therefore\ T_{predicted,out} = 18.66\ °C$$

$$T_{measured} = 19.1\ °C$$

$$\Delta T =\ 19.1\ °C -\ 18.66\ °C$$

$$\therefore\ \Delta T = 0.44\ °C$$

The rate of heat transfer from the water to the air in the cooling tower was calculated using enthalpy balances. Two different heat transfer equations were solved. The liquid water stream heat transfer, $Q_{predicted,w}$, represents the thermal energy lost or gained by the water stream. The other equation was the moist air stream heat transfer, $Q_{predicted,a}$, which represents the energy gained by air. The water stream tracks energy in and out of liquid water and the air stream uses the enthalpy of moist air. They both represent energy balance on the water side and the moist air side. Ideally, these two values should be the same because of energy conservation, meaning the heat lost by the water should be the same as the heat gained by the moist air. The liquid water stream heat transfer is dependent on the mass flowrate of the water entering the tower, $\dot{m}_{w,in}$, the mass flowrate of the water leaving the cooling tower, $\dot{m}_{w,out}$, enthalpy of the incoming water, $h_{w,in}$, and the specific enthalpy of the water leaving the cooling tower, $h_{w,out}$. The air stream heat transfer depends on mass flowrate of the dry air, $\dot{m}_{air}$, the enthalpy of the air entering the tower, h_{in}, and the enthalpy of the air leaving the tower, h_{out}.

$$Q_{predicted,w} = \dot{m}_{w,in} * h_{w,in} - \dot{m}_{w,out} * h_{w,out}\ p.8\ Schulze\ [3] \tag{9}$$

$$Q_{predicted,a} = \dot{m}_{air} * (h_{out} - h_{in})\ p.8\ Schulze\ [3] \tag{10}$$

$$Q_{predicted,w} = 0.03\frac{\cancel{kg}}{s} * 96.33\frac{kJ}{\cancel{kg}} - 0.0297\frac{\cancel{kg}}{s} * 78.22\frac{kJ}{\cancel{kg}}$$

$$\therefore Q_{predicted,w} = 0.57\ kW$$

$$Q_{predicted,a} = 0.083\frac{\cancel{kg}}{s} * \left(71.9\frac{kJ}{\cancel{kg}} - 65\frac{kJ}{\cancel{kg}}\right)$$

$$\therefore Q_{predicted,a} = 0.57\ kW$$

The actual heat transfer in the cooling tower, $Q_{measured}$, was determined from taking the total power supplied to the system, Q_{total}, and then subtracting the portion of the power that goes to the damper, Q_{damper}, equal to a constant 0.162 kW. This subtraction of energy losses removes wasted energy and only leaves the useful heat that transfers from the water to the air. Furthermore, a difference in heat transfer, ΔQ, can be found by subtracting the predicted heat transfer value, $Q_{predicted}$, from the actual heat transfer in the cooling tower, $Q_{measured}$, to explain how accurate the model is and show the amount of error or heat loss in the system.

$$Q_{measured} = Q_{total} - Q_{damper}\ Eqn.1.12c, p.15\ Bergman\ et\ al.\ [1] \tag{11}$$

$$\Delta Q = Q_{measured} - Q_{predicted}\ p.12\ Schulze\ [7] \tag{12}$$

$$Q_{measured} = 0.532\ kW - 0.162\ kW$$

$$\therefore Q_{measured} = 0.37\ kW$$

$$\Delta Q = 0.37\ kW - 0.57\ kW$$

$$\therefore \Delta Q = -0.20\ kW$$

Table 3: The table below shows the original data from the experiment for each test case at different heating power inputs and water flow rates where ΔP_{air} is the pressure drop of the air stream, T_1 is the inlet dry bulb temperature, T_2 is the inlet wet bulb temperature, T_3 is the outlet dry bulb temperature, T_4 is the outlet wet bulb temperature, T_5 is the inlet water temperature, and T_6 is the outlet water temperature. The first three test cases involved constant low heater power input at 0.37 kW, while the final three test cases experienced increased heater power input at 1.228 kW. For both the low heat and high heat test cases, the water inlet flow rate was set to 20 g/s, 30 g/s and 40 g/s. The pressure drop of the air stream, measured using a manometer across the orifice, was set to a constant 30 mmH_2O. The inlet dry and wet bulb temperatures remained nearly constant, with T_1 ranging from 20.4 °C to 20.9 °C and T_2 from 16.0 °C to 16.5 °C, indicating stable ambient conditions. In contrast, the outlet dry and wet bulb temperatures increased with both increasing power and water flow rate, with T_3 rising from 18.6-19.3 °C at low heating power to 21.5-23.6 °C at high power, while T_4 approached T_3 more closely at higher temperatures, indicating that the air became increasingly saturated. The water temperatures showed the strongest dependence on operating conditions, as the inlet water temperatures increased significantly with heater input from approximately 22.7-23.3 °C to 31.8-33.2 °C., and the outlet water temperature increased from 18.5-19.3 °C to 20.6-23.0 °C. In all cases, T_6 remained lower than T_5, confirming effective cooling through evaporation. Additionally, increasing water flow rate at constant heating power resulted in slightly higher outlet water temperatures due to the distribution of energy over a larger mass flow. Overall, these trends were consistent with expected cooling tower behavior, demonstrating increased evaporation and heat transfer with higher thermal input.

Test	**Heating Power (kW)**	**Water Flow Rate (g/s)**	**ΔP_{air} (mmH_2O)**	**T_1 (°C)**	**T_2 (°C)**	**T_3 (°C)**	**T_4 (°C)**	**T_5 (°C)**	**T_6 (°C)**
1	0.37	20	30	20.4	16.0	18.6	18.3	23.3	18.5
2	0.37	30	30	20.6	16.0	19.1	18.9	23.0	19.1
3	0.37	40	30	20.6	16.1	19.3	19.1	22.7	19.3
4	1.228	20	30	20.7	16.2	21.5	21.6	33.2	20.6
5	1.228	30	30	20.7	16.4	22.9	23.0	32.8	21.2
6	1.228	40	30	20.9	16.5	23.6	23.7	31.8	23.0

References:

[1] T. L. Bergman, A. S. Lavine, F. P. Incropera, and D. P. DeWitt, *Fundamentals of Heat and Mass Transfer*, 8th ed. Hoboken, NJ, USA: Wiley, 2017, pp. 15, 72
[2] K. Schulze, *Cooling Tower 1*. Gainesville, FL: 2026, p. 18
[3] K. Schulze, *Cooling Tower 3*. Gainesville, FL: 2026, pp. 7-12
[4] R. W. Fox and J. W. Mitchell, *Fox and McDonald's Introduction to Fluid Mechanics*, 10th ed. Hoboken, NJ, USA: Wiley, 2019
[5] K. Schulze, *Cooling Tower 4*. Gainesville, FL: 2026, p. 21
[6] K. Schulze, *Cooling Tower 4*. Gainesville, FL: 2026, p. 27
[7] K. Schulze, *Cooling Tower 5*. Gainesville, FL: 2026, p. 10-12

Table 4: Lab Report Participation Log

Name	Date	Hours Worked	Description of Tasks Performed
Bryce Addeo	3/18/2026	1	Report formatting and 1st paragraph cover letter
Sean Muller	3/28/2026	2	Calculation of relative humidity and humidity ratio
Bryce Addeo	3/28/2026	3	Calculation of relative humidity and humidity ratio
Bryce Addeo	3/29/2026	1.5	Calculation of relative humidity and humidity ratio and filling in tables with results
Matthew Strogach	3/31/2026	2.5	H,in and h,out determinations, hw,in calculation
Bryce Addeo	3/31/2026	3.5	Completed all calculations to solve ΔT & ΔQ
Bryce Addeo	4/1/2026	3	Completed all cover letter paragraphs
Bryce Addeo	4/3/2026	3	Wrote descriptions for Tables 1, 2 and 3
Matthew Strogach	4/3/2026	4	Sample calcs and descriptions for rho(air), rho(mix), mdot,air, mdot,w,out, h,w,out, T(predicted), deltaT
Bryce Addeo	4/4/2026	1	Proofread/editing cover letter and sample calcs
Matthew Strogach	4/5/2026	1.5	Descriptions for sample calcs of Qpredicted,w, Qpredicted,a, Qmeasured, and deltaQ
Sean Muller	4/5/2026	6	Sample calcs and descriptions for y, omega low, phi low, omega high, phi high, omega in, phi in, explanation of process for omega out and phi out, Q predicted w, Q predicted a, Q measured, Delta Q
Sean Muller	4/5/2026	2	Sources for sample calculations
Bryce Addeo	4/5/2026	0.5	Final proofread and edits
	Total Hours	34.5	

Total Hours per Group Member: Bryce: 16.5 hrs, Matt: 8 hrs, Sean: 10 hrs